

Shervin Faramarzi Babadi

Emails: faramarzi.shervin@gmail.com, shervin.faramarzi.babadi@smail.th-koeln.de

Mobile: +49 (1) 5238 703395

Address: Cologne, Germany

EDUCATION

- **M.Sc. , TH Köln University of Applied Sciences** Cologne, Germany
Master of Science in Automation & Information Technology (IT) 2026 – Present
- **M.Sc., Azad University Najafabad Branch** Isfahan, Iran
Master of Science in Electrical Engineering (Control System) 2015 – 2019
Thesis Title: Adaptive Fuzzy Iterative Learning Control for Quadrotor Path Tracking
cGPA: 17.43 (out of 20)
Advisor: Dr. Abbas chatraei
- **B.Sc., Azad University Najafabad Branch** Isfahan, Iran
Bachelor of Science in Electrical Engineering (Control System) 2011 – 2015
Thesis Title: Performance of Static Var Compensation Systems (SVC) in Steel Companies
cGPA: 16.05 (out of 20)
Advisor: Dr. Homayoun Mahdavi-Nasab

FIELDS OF INTERESTS

- **Machine Learning:** Deep Learning, Artificial Intelligence, Neural Networks
- **Robotics:** Industrial Robots, Collaborative Robots, Autonomous Mobile Robots (AMRs), Human-Robot Collaboration (HRC), Tele-robotic Systems
- **Controllers:** Optimal, Fuzzy, and Adaptive Control of robotic and mechatronic systems.

PUBLICATIONS

- P1 **Shervin. Faramarzi**, Abbas Chatraei, “Adaptive Fuzzy Iterative Learning Control for Quadrotor Path Tracking, *5th National Conference on Electrical Engineering of Iran*, Najafabad, Iran, 2018.
- P2 **Shervin. Faramarzi**, Abbas Chatraei, “Design of Adaptive Fuzzy Iterative Learning Control with Forgetting Factor for Quadrotor Path Tracking, *Transactions of the Institute of Measurement and Control*, Submitted.

ACADEMIC EXPERIENCE

- **Azad University Najafabad Branch, Isfahan, Iran** Fall 2013
Teaching Assistant — Linear Control Systems
Assisted in teaching classical control methods (root locus, Bode plots, PID) using MATLAB/Simulink, Supported students in modeling and analysis of dynamic system
- **Azad University Najafabad Branch, Isfahan, Iran** Fall 2017
Teaching Assistant — System Identification
Assisted in lectures and labs on parameter estimation, model validation, and identification algorithms using MATLAB/Simulink; evaluated assignments and provided feedback.

EMPLOYMENT

- **Makan System Daneshmand Company, Isfahan, Iran** Aug 2023 – Jan 2026
Programmer & Data Analyst
Developed Python-based data processing workflows for unstructured technical data and applied data-driven analysis using pandapower to perform load flow analysis and network simulation for evaluating the performance and reliability of electrical power systems.
- **Atiehsazanfarhood Company, Isfahan, Iran** Nov 2019 – Aug 2023
Research & Development Engineer
Worked on control system design and optimization for medical devices in safety-critical environments, applying system modeling and analysis to improve stability, performance, and robustness under operational constraints.
- **Azad University Najafabad Branch, Robotics Institution, Isfahan, Iran** 2018 – 2019
Part-time Researcher
Modeled and simulated a SCARA robot arm, implemented control algorithms for trajectory tracking, and evaluated system performance through simulation-based analysis.
- **Mobarakeh Steel Company, Mobarakeh, Isfahan, Iran** Feb 2015 – Apr 2015
Intern
Worked with PLC-based automation systems in industrial power distribution, assisting in troubleshooting and gaining hands-on experience with large-scale electrical and automation systems.

PROJECTS

- **Modeling and Simulation of Quadrotor:** Developed a nonlinear dynamic model of a quadrotor and implemented control strategies in MATLAB/Simulink to evaluate trajectory tracking performance under varying conditions.
- **Modeling and Simulation of Robot Arm:** Modeled the kinematics and dynamics of a SCARA robotic manipulator and implemented trajectory tracking controllers in MATLAB, evaluating system performance through simulation.
- **Inverted Pendulum LQR Control:** Derived system dynamics using Lagrange equations and designed an LQR controller in MATLAB to achieve stabilization, analyzing system response and control performance.

AWARDS

- Ranked 1st among M.Sc. students in Control Systems, Electrical Engineering Department, Azad University Najafabad.
- Ranked 2nd among B.Sc. students in Control Systems, Electrical Engineering Department, Azad University Najafabad.

TECHNICAL SKILLS

- **Control Systems:** Adaptive Control, Fuzzy Control, LQR, System Modeling, and Simulation
- **Power Systems:** Load Flow Analysis, Network Simulation (pandapower)
- **Optimization:** Numerical Optimization Methods
- **Data Preparation:** cleaning and structuring text data from notebook sources and report preparation
- **Software:** Git, Microsoft Word, Excel

PROGRAMMING AND SOFTWARE SKILLS

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| • Python | • MATLAB |
| • Git | • C++ |

CERTIFICATES & PROFESSIONAL TRAINING

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|---|----------|
| • Capstone: Retrieving, Processing, and Visualizing Data with Python
<i>University of Michigan — Coursera</i> | Jan 2026 |
| • Using Python to Access Web Data
<i>University of Michigan — Coursera</i> | Sep 2025 |
| • Programming for Everybody (Getting Started with Python)
<i>University of Michigan — Coursera</i> | Jun 2025 |
| • Python Data Structures
<i>University of Michigan — Coursera</i> | Aug 2025 |

LANGUAGE PROFICIENCY

- **English:** Fluent C1 *IELTS Overall Score: 7.0 (C1) (Speaking: 7.0, Listening: 8.5, Reading: 7.0, Writing: 6.0)*
- **Deutsch:** A2
- **Farsi (Persian):** Native

EXTRACURRICULAR ACTIVITIES

- Swimming, Tennis, Aerobics, Yoga

REFERENCES AVAILABLE TO CONTACT

- **Abbas Chatraei, Ph.D.**
*Assistant Professor of Electrical Engineering Department
Azad University Najafabad Branch, Isfahan, Iran
Email: abbas.chatraei@pel.iaun.ac.ir*
- **Mahdi S. Hosseini, Ph.D.**
*Assistant Professor of Computer Science and Software Engineering Department
Concordia University, Montreal, Canada
Email: mahdi.hosseini@concordia.ca*
- **Amirhosein Adami, Ph.D**
*CEO & Managing Director
Makan System Daneshmand, Isfahan, Iran
Email: amirhoseinadami@gmail.com*